Quantum Field Theory I

First Homework Assignment

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Problem

We consider a four-dimensional hypercubic lattice with lattice spacing Δ. The lattice propagator is written as a sum over all lattice paths from 0 to x, where each link contributes a factor $e^{-m_0\Delta}$. The directions of each link are

$$\nu_k \in \{(\pm 1,0,0,0),(0,\pm 1,0,0),(0,0,\pm 1,0),(0,0,0,\pm 1)\}\,.$$

The length of a path containing n links is

$$L = n\Delta.$$

Therefore, the path sum can be written as

$$A(x) = \sum_{n=0}^{\infty} e^{-m_0\Delta n} \sum_{\{\nu_k\}} \delta^{(4)}_{\sum_{k=1}^{n}\nu_k,\, x/\Delta}. \tag{1}$$

Here the arguments of the Kronecker delta are dimensionless, since x/Δ is the dimensionless lattice coordinate.

Fourier Representation of the Kronecker Delta

For dimensionless integer-valued four-vectors, the Kronecker delta can be represented as

$$\delta^{(4)}_{\boldsymbol{N},\boldsymbol{M}} = \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4} e^{ik\cdot(\boldsymbol{N}-\boldsymbol{M})}. \tag{2}$$

Applying this to Eq. (1), with

$$\boldsymbol{N} = \sum_{k=1}^{n} \nu_k, \qquad \boldsymbol{M} = \frac{x}{\Delta},$$

gives

$$\delta^{(4)}_{\sum_{k=1}^{n}\nu_k,\, x/\Delta} = \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4} \exp\left[ik\cdot\left(\sum_{k=1}^{n}\nu_k - \frac{x}{\Delta}\right)\right]. \tag{3}$$

Thus,

$$A(x) = \sum_{n=0}^{\infty} e^{-m_0\Delta n} \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4} e^{-ik\cdot x/\Delta} \prod_{j=1}^{n}\left(\sum_{\nu_j} e^{ik\cdot\nu_j}\right). \tag{4}$$

Since the four-dimensional hypercubic lattice has eight possible directions, we have

$$\sum_{\nu} e^{ik\cdot\nu} = 2\left[\cos k_1 + \cos k_2 + \cos k_3 + \cos k_4\right]. \tag{5}$$

Therefore,

$$A(x) = \sum_{n=0}^{\infty} e^{-m_0\Delta n} \int_{-\pi}^{\pi} \frac{d^4k}{(2\pi)^4} e^{-ik\cdot x/\Delta} \left[2\sum_{\mu=1}^{4} \cos k_\mu\right]^n. \tag{6}$$

Now k is a *dimensionless* lattice momentum. To introduce the physical momentum p, we define

$$k_\mu = p_\mu \Delta, \qquad p_\mu = \frac{k_\mu}{\Delta}. \tag{7}$$

Consequently, the integration range becomes

$$-\frac{\pi}{\Delta} \le p_\mu \le \frac{\pi}{\Delta},$$

and the exponent becomes

$$e^{-ik\cdot x/\Delta} = e^{-ip\cdot x}.$$

Thus, up to the overall normalization factor coming from $d^4k = \Delta^4 d^4p$, we can write

$$A(x) \sim \sum_{n=0}^{\infty} e^{-m_0\Delta n} \int d^4p\, e^{-ip\cdot x} \left[2\sum_{\mu=1}^{4} \cos(p_\mu\Delta)\right]^n. \tag{8}$$

For small Δ,

$$2\sum_{\mu=1}^{4} \cos(p_\mu\Delta) = 8 - p^2\Delta^2 + O(\Delta^4), \tag{9}$$

where

$$p^2 = p_1^2 + p_2^2 + p_3^2 + p_4^2.$$

Hence,

$$\begin{aligned} \left[2\sum_{\mu=1}^{4} \cos(p_\mu\Delta)\right]^n &\simeq \left(8 - p^2\Delta^2\right)^n \\ &= 8^n \left(1 - \frac{p^2\Delta^2}{8}\right)^n \\ &\simeq 8^n e^{-np^2\Delta^2/8}. \end{aligned} \tag{10}$$

Substituting this into Eq. (8),

$$A(x) \sim \sum_{n=0}^{\infty} e^{-m_0\Delta n} 8^n \int d^4p\, e^{-np^2\Delta^2/8} e^{-ip\cdot x}. \tag{11}$$

Therefore,

$$A(x) \sim \sum_{n=0}^{\infty} \exp\left[-n\left(m_0\Delta - \ln 8\right)\right] \int d^4p\, e^{-np^2\Delta^2/8} e^{-ip\cdot x}. \tag{12}$$

We see that the important combination of Δ and n is $n\Delta^2$, rather than $n\Delta$. Thus, in the continuum limit, we define

$$s = \frac{n\Delta^2}{8}. \tag{13}$$

In order for long paths to remain relevant in the continuum limit, the factor multiplying n in Eq. (12) must be of order Δ^2. Therefore, we tune the bare mass according to

$$m_0\Delta - \ln 8 = \frac{m^2\Delta^2}{8}. \tag{14}$$

This gives

$$\boxed{m_0(\Delta) = \frac{\ln 8}{\Delta} + \frac{m^2}{8}\Delta}. \tag{15}$$

With this choice,

$$e^{-n(m_0\Delta - \ln 8)} = e^{-m^2 n\Delta^2/8} = e^{-m^2 s}, \tag{16}$$

and therefore

$$A(x) \sim \sum_n e^{-m^2 s} \int d^4p\, e^{-sp^2} e^{-ip\cdot x}. \tag{17}$$

In the continuum limit, the sum over n becomes an integral over s, up to an overall normalization factor:

$$A(x) \sim \int_0^\infty ds\, e^{-m^2 s} \int d^4p\, e^{-sp^2} e^{-ip\cdot x}. \tag{18}$$

Thus the required continuum limit is obtained by tuning the bare mass as in Eq. (15).

Shortest Path Domination for Finite Δ

For finite Δ, there is another regime in which the shortest paths dominate the path sum.

Each additional link has a factor

$$e^{-m_0\Delta},$$

while at every step there are eight possible directions. Thus, very roughly, the competition between the number of paths and the suppression of each path is controlled by

$$8e^{-m_0\Delta}. \tag{19}$$

If

$$8e^{-m_0\Delta} < 1, \tag{20}$$

or equivalently

$$m_0\Delta > \ln 8, \tag{21}$$

longer paths are exponentially suppressed.

For a fixed endpoint x, the shortest possible path has the minimum number of links,

$$n_{\min} = \sum_{\mu=1}^{4} \left| \frac{x_\mu}{\Delta} \right|. \tag{22}$$

Any longer path must contain additional steps. In particular, because the lattice is bipartite, paths connecting the same two points differ in length by an even number of links, so a path with length

$$n = n_{\min} + 2r$$

contains $2r$ additional links compared with a shortest path.

Each additional pair of links introduces an exponential suppression of the form

$$e^{-2m_0\Delta}.$$

Therefore, when

$$m_0\Delta \gg \ln 8,$$

the exponential suppression of longer paths overwhelms the growth in the number of available paths, and the path sum is dominated by the shortest paths.

Hence, for finite Δ, the two relevant regimes can be summarized as

$$\begin{aligned} m_0\Delta > \ln 8 \quad &\Longrightarrow \quad \text{long paths are suppressed,} \\ m_0\Delta \simeq \ln 8 \quad &\Longrightarrow \quad \text{long paths can remain relevant.} \end{aligned} \tag{23}$$

In particular, the continuum limit requires tuning $m_0\Delta$ to approach $\ln 8$ with a difference of order Δ^2, as in Eq. (14).

For a fixed finite m_0 while taking $\Delta \to 0$, however,

$$m_0\Delta \longrightarrow 0,$$

and therefore

$$8e^{-m_0\Delta} \longrightarrow 8 > 1.$$

Thus the path sum is not in the shortest-path regime in this limit; instead, the number of long paths grows too rapidly and the sum diverges. The shortest-path domination discussed above applies to finite Δ when $m_0\Delta$ is sufficiently larger than $\ln 8$.

AI use declaration

The AI model used in preparing and revising this homework solution was **GPT-5.6 Luna**.

The solution was checked using the course lecture notes as a reference, in particular the derivation of the lattice path sum, the Fourier representation, and the continuum limit involving the combination $n\Delta^2$.

The AI was also used to revise and correct a small mistake in the calculation: the Kronecker delta was rewritten with dimensionless arguments, and the distinction between the dimensionless lattice momentum k and the physical momentum p was made explicit. The main derivation and notation of the original solution were otherwise retained.
The AI was also used to convert the hand written solution into Latex code.